

INTERMEDIATE REPORT – ASSIGNMENT 4

COMPUTATIONAL BIOFLUID MECHANICS
ACADEMIC YEAR 2025-2026

Name student

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ASSIGNMENT 4: STEADY-STATE SIMULATION OF POST-OP CASE

Make sure that figures/tables are high-quality and complete, that they have numbers and clear captions, and that they are referred to in the text.

Methods: Meshing (/2)

- *Provide a table with the relevant mesh parameters, regarding element size, number and type. (/1). A measure of a good table is that a different researcher would be able to replicate your mesh with the included information.*
- *Can you explain why we use different elements near the boundaries? (/1) (max. 100 words)*

Methods: Simulation set-up (/4)

- *Describe the simulation set-up: mention the relevant material properties of the fluid (/0.5) and the boundary conditions (/2) (max. 100 words). Use references where needed.*
- *Include a figure of the residuals. Explain how you assessed convergence and whether you think this simulation is converged (/1) (max. 100 words).*
- *Suppose that a parameter of interest is the velocity in the axial direction, and that you calculate it based on your current solution. If you add 1000 iterations to your simulation, what would the impact be on that value? (/0.5) (max. 50 words)*

Results (/3)

- *Visualize the flow and pressure behavior. What is the flow distribution over the left and right lung? (/1) Can you explain why the fluid distributes (un)equally? (/1) (max. 100 words)*
- *Use CFD results to calculate the local resistance over a specific section before the bifurcation, similar to where the constriction is located in the pre-op case (indicate visually). Show clearly which formula you use (/1) (max. 100 words).*

Future work (/1)

For the mesh sensitivity analysis, which parameters would you focus on? Of course, you may be inspired by the previous questions. (max. 100 words)